

ARMA IDENTIFICATION: A SURVEY

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Abstract : This paper presents a survey of the main ARMA parameter estimation methods

Keywords : Parameter estimation, AutoRegressive Moving-Average, non-gaussian, non-minimum phase systems, high-order statistics.

INTRODUCTION

ARMA models are very useful for Signal Processing and Automatic Control applications like parametric representation of signals and systems, spectral estimation and adaptive prediction of stochastic processes (Puy, Favier, Siffredi, 1989; TFavier, Dubois, 1990).

The purpose of this paper is to survey the main methods for ARMA(p,q) model identification, based on second-order statistics or higher-order statistics.

During the last decade, many methods were proposed for identifying ARMA models. These methods can be classified in different ways :

- C1 : According to the assumption on the input process (i.i.d gaussian or non-gaussian process) or the characteristics of the ARMA model to be identified (causal or noncausal, minimum phase or nonminimum phase ARMA models) :

- . C11 : Methods based on second-order statistics.
- . C12 : Methods based on higher-order statistics (third-order or fourth-order cumulants).

- C2 : According to the information (measured data or sampled covariance or cumulant sequences) and the data processing type which are used :

- . C21 : Methods based on a sequential processing of the measurements (time recursive methods). Recursive methods can be subdivided into *adaptive* methods (for nonstationary ARMA processes with time-varying parameters) and *non-adaptive* methods (for stationary ARMA processes).
- . C22 : Methods based on a block processing of the measurements (non recursive or batch-type methods).

Up to now, the methods based on higher-order statistics (class C12) are essentially non-recursive.

- C3 : According to the estimation procedure which is used :

- . C31 : Methods based on a one-step procedure, with a simultaneous estimation of the AR and MA parameters.

- . C32 : Methods based on a two-steps procedure, with a separated estimation of the AR parameters and then the MA parameters.

We have to notice that for certain applications (adaptive control, spectral estimation, ...) the estimation of the MA part can be unnecessary or replaced by the estimation of other parameters like for example the numerator spectral coefficients (Moses, Beex, 1986).

- C4 : According to the assumption on the additive noise (independent of the input and output signals, white or colored, gaussian or non-gaussian, bounded or unbounded).
When the additive noise is bounded and the bounds on its instantaneous values are known, robust estimation methods can be used (Arruda, Favier, 1991; Rao, Huang, Dasgupta, 1990).

In order to improve the computational efficiency (in terms of computation time), most of the recent ARMA identification methods make use of the techniques of fast algorithms (lattice filters, Chandrasekhar equations).

Among the methods of the class C11(second-order statistics), we first consider the methods based on a one-step procedure (class C31) :

- The Maximum Likelihood method which needs to solve a system of nonlinear equations (Newbold, 1974; Anderson, 1977; Dugre and co-workers, 1986), and its numerous approximated versions like RML1, RML2 and modified RML2 method (Panuska, 1968; Young, 1968; Söderstrom, 1973; Friedlander, 1982c).

- Methods based on spectral matching technique (Friedlander, Porat, 1984) which involves nonlinear optimization and can be viewed as an approximated form of the Maximum Likelihood method.

- Methods based upon lattice and ladder realizations obtained from the decomposition of the ARMA process into a two dimensional AR process, and connected with the classical Extended Least Squares procedure. These algorithms were proposed by Morf and Lee (1979), and in a normalized form by Lee and co-workers (1980) in the case of an ARMA(p,p) model,

then by Benveniste and Chaure (1981) and Friedlander (1983b) in the case of an ARMA(p,q) model.

We now consider, for the class C11, methods based on a two-steps procedure (C32) and allowing to estimate the AR parameters :

- Methods based on the so-called Modified (or Extended) Yule - Walker (MYW) equations. In (Gersch, 1970), these equations, taken in a minimal number (q), involve the sample correlation coefficients of the ARMA process. In (Chow, 1972) it is suggested to use a number of equations that is greater to q in order to improve the estimation accuracy, which leads to an overdetermined set of MYW equations. This idea was used by Cadzow and Moses (1981) in the context of spectral estimation (see also Cadzow (1982)).

Another way to improve this method consists in combining it with a maximum likelihood approach (Stoica and co-workers, 1987a).

In (Favier, 1977a, 1977b) the MYW equations are expressed in terms of the autocorrelation function of a pseudo innovation process or the crosscorrelation function between the ARMA process and a pseudo innovation process.

The MYW equations are strongly linked with the Instrumental Variable method (Friedlander, 1983a), and an asymptotic analysis of this kind of method is carried out in (Stoica and co-workers, 1985, 1987b, 1987c). Finally these equations can be solved in a recursive way (class C21) in using the techniques of lattice filters (Cadzow, Moses, 1981; Moses and co-workers, 1985; Samson, 1982).

- Methods based on an iterated least-squares procedure that yields consistent least squares estimates for the AR parameters, which are asymptotically equivalent to the solution of the MYW equations in the stationary case (Tsay, Tiao, 1984).

A recursive version (class C21) of this algorithm was derived by using a least-squares lattice whitening filter of order (p+q) and then solving a set of q linear equations (Li, Dickinson, 1986).

MA parameter estimates can also be obtained using this approach by simply normalizing the auxiliary variables which are solutions of this set of linear equations (Li, Dickinson, 1988). Finally, a dual version of this algorithm was proposed recently by (Favier, Puy, Maynard, 1989), (Puy, 1990) and extended to the case of nonstationary ARMA models by (Settineri, 1991).

Other methods allow to estimate both the AR and MA parameters in a two-steps procedure (class C32):

- Methods based on the identification of long AR models (Durbin, 1959, 1960; Graupe and co-workers, 1975) or long MA models (Favier, 1982a, 1982b). In these two last references the duality between these two approaches (long AR and long MA approximations) is clearly pointed out. This class of methods will be called dual methods.

With regard to the estimation methods of the MA part, also called spectral factorization methods, two approaches using estimated correlation functions are possible:

- Methods based on a Cholesky factorization (Morf, 1974; Wilson, 1969, 1972), with the possibility to use a lattice algorithm to make the calculations (Friedlander, 1983c).

- Stochastic realization algorithms needing to solve Riccati or Chandrasekhar type equations (Dickinson and co-workers, 1974; Favier, 1977, 1979, 1982b; Favier, Alengrin, 1979; Favier, Salut, 1979).

We now consider the methods based on higher-order statistics (class C12). The early works in this field are due to (Brillinger, 1965), (Brillinger, Rosenblatt, 1967).

In the class C31 (one-step estimation procedure), we distinguish :

- Methods based on cumulant matching techniques (Tugnait, 1987), (Tugnait, 1990b), (Giannakis, Swami, 1990) the philosophy of which is the same as that of (Friedlander, Porat, 1984), leading to a nonlinear optimization.

- Methods based on the computation of the complex cepstrum of higher-order cumulants (Nikias, Pan, 1987), (Pan, Nikias, 1988) working for non-minimum phase ARMA models.

Looking into the class C32 (two-steps estimation procedure), the AR part of the model is estimated by use of the YWM equations extended to the third or fourth-order statistics, that is to say the autocorrelation is replaced by the third or fourth-order cumulants (Giannakis, Mendel, 1989 and 1990), (Tugnait, 1988 and 1989).

With regard to the estimation methods of the MA part, there are mainly three approaches :

- Methods based on the so-called "residual" process (or "AR-compensated" process) obtained after filtering the ARMA process through the AR(p) estimated part. Once this AR part has been estimated, a MA estimation method is applied to the AR-compensated process .

The "C(q,k)" algorithm, initially proposed by (Giannakis, 1987), allows to compute in a simple way the estimated parameters as a function of the third-order cumulants of the MA process ("AR-compensated" process). An extension to the fourth-order cumulants was derived by (Dianat, Raghuveer, 1989).

Another approach consists in using both second-order and higher-order cumulants in the estimation procedure, leading to "Cumulant and Correlation based" methods.

(Giannakis, Mendel, 1989) proposed a method based on the least-squares solution of an overdetermined set of $(3q+1)$ linear equations with $(2q+1)$ unknown terms. Different strategies are used to determine the q MA parameters in function of the $(2q+1)$ redundant terms (the q MA parameters, their square values and the ratio ϵ between the second and third-order input cumulants of the ARMA model).

An improvement of this method (Tugnait, 1990a) consists in deriving an overdetermined set of $4q$ linear equations with $(3q+2)$ unknown terms, in such a way that the pseudo-inverse gives a unique solution.

Furthermore, to avoid the redundancy appearing in the vector containing the unknown terms, Tugnait proposed in the same paper to solve a set of $2q$ linear equations with $(q+1)$ unknown coefficients (the q MA parameters and the ratio ϵ).

Finally, (Porat, Friedlander, 1989) derived a general lower bound for the variance of estimates based on high-order sample moments. This lower bound is used to analyse the performance of the (Giannakis, Mendel, 1989) method.

- Methods based on the determination of the impulse response of the ARMA model.

For this purpose, (Swami, Mendel - 1990) proposed the following two-steps procedure denoted "q-slice" algorithm, for estimating the MA parameters :

1st step: the impulse response of the ARMA model is computed as a function of the AR estimated parameters and the third-order output cumulants,

2nd step: the MA estimated parameters are computed as a function of the impulse response and the AR estimated parameters.

We have to note that the AR part is estimated by use of YWM "extended" equations, as we have shown early.

- Another approach proposed by (Giannakis, Swami, 1990) works for the case of causal as well as non-causal ARMA models. Indeed, in the latter case, the use of the YWM "extended" equations is prohibited. The method is based on the decomposition of the bispectrum of the ARMA process into two terms which can be viewed respectively as the bispectrum of two MA processes of order p and q , the parameters of which are the AR and MA coefficients of the ARMA model. These coefficients are estimated by applying twice the "C(q,k)" algorithm of (Giannakis, 1987) or a numerically better version of this algorithm (Giannakis, Swami, 1990).

Finally, there are very few recursive algorithms based on high-order statistics, most of those proposed using lattice algorithms. In (Swami, Mendel - 1988a), (Swami, Mendel - 1988b), the time-update of the AR part is made by use of a "double" lattice filter, and a recursive version of the "C(q,k)" algorithm of (Giannakis, 1987) is used for the MA part. In (Friedlander, Porat, 1989), the AR part is computed in a recursive way by use of a "classical" - i.e. based on second-order statistics - lattice filter. For the MA part, a recursive version of the method of (Giannakis, Mendel, 1989) based on the ORIV (Overdetermined Recursive Instrumental Variable) algorithm (Friedlander, 1984) is used.

We have to note that there exists two important papers dealing with estimation methods based on higher-order statistics: the first one due to (Nikias, Raghuveer, 1987) where all the basic results concerning cumulants and polyspectra are derived; the second one due to (Mendel, 1991) where a synthetic presentation of AR, MA and ARMA parameter estimation methods is made.

An extended version of this paper is in course of preparation and will be sent on request.

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